ALIYA ALSAFA

COMPUTER SCIENCE

CONTACT

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Location: Palo Alto, CA; New York, NY; Chicago , IL

SKILLS

- **Languages:** C/C++, Python, Java, Rust, Terraform, SQL
- **Dev Tools:** Git, Jira, Wiz, Sumo Logic, DoControl, BlinkOps, Wireshark, VSCode, Vim, Intellij, PyCharm
- **Platforms:** AWS Cloud, Docker, Linux/Unix, Windows

PROJECTS

- Implemented an end-to-end **TCP protocol** from scratch with bytestream, sender/receiver, and retransmission logic.
- Contributing research toward a **CoNEXT 2026** paper on network flow offload with commodity NICs (Stanford ESRG).
- Participated in **Sekai CTF 2024** competition.

ACHIEVEMENTS

- Lang Lang International Music Foundation Young Scholar (Aug 2018 - Present)
    - Nonprofit organization that provides opportunities to perform internationally at concert halls, schools, hospitals, senior centers, and more

EXTRACURRICULARS

- Member, Women in Computer Science
- Member, Stanford Applied Cyber
- Executive Board, Stanford Piano Society
- Executive Board, Stanford XTRM Dance
- Member, Legacy Dance

WORK EXPERIENCE

Empirical Security Research Group (ESRG)

Research Assistant — Summer 2025

- Built Rust modules in the Retina network analysis framework to offload flow-rule installation onto commodity NICs (DPDK, Mellanox).
- Designed experiments to measure NIC performance (rule latency, drop counters, RSS distribution) under high-throughput workloads.
- Presented findings in research poster “Dropping Traffic on Commodity NICs” at Stanford ESRG summer session

Redis

Security Engineering Intern — Summer 2024

- Penetration Testing
    - Evaluated HackerOne bug bounty reports and attempted to reproduce findings using penetration testing techniques
    - Developed a comprehensive guide on offensive testing to train software developers in conducting their own penetration tests
- Remediation and report of orphaned domains
    - Created a tool to find and report on potentially abandoned corporate DNS records to help reduce cloud attack surface
    - Created, containerized, and deployed tool as a python script to AWS ECS cluster
- Hardening Baselines Initiative
    - Created custom frameworks within Wiz to scan the cloud environment for vulnerabilities in compliance with the CIS Ubuntu Benchmarks
- Assess and Improve Control Automation
    - Integrated data security tool DoControl into corporate SaaS apps to empower control owners to implement effective protections

EDUCATION

Stanford University — California

Computer Science Major, B.S.

Piano Performance, Music Minor — Anticipated Graduation, 2026

- GPA: 3.65
- Relevant Coursework:
    - CS 111 (Operating Systems), CS 107 (Computer Organization and Systems), CS 144 (Networking), CS, 155 (Cybersecurity), CS 161 (Algorithms), CS 103 (Mathematical Foundations of Computing), CS 109 (Probability for Computer Science), CS 80E (Computer Architecture), MATH 51 (Linear Algebra)

Juilliard Pre-College — New York

Diploma — 2015 - 2022

- Trained as a classical concert pianist for 7 years with the top piano performance faculty in the nation. Participated in private lessons, chamber music, and high-level music electives every Saturday.

Special Music School High School — New York

High School Diploma — 2018 - 2022

- ACT Score: 36
- Valedictorian

Harvard Summer School — 2021

- CSCI-7 Introduction to Computer Programming with Python